Vortical Enclosure Dynamics, Vol. 1

Foundations

Independent Research — 2026

Yuhki Endoh

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Contents

1 Introduction **3**

2 The Minimal Axiom **3**
2.1 From Difference to Dynamics 4
2.2 The Non-Closure Constraint 5

3 Causal Logs and Local Time-Like Accumulation **5**
3.1 History and Structure 5
3.2 The Self-Referential Selection Equation 6
3.3 Local Time-Like Accumulation 6
3.4 The Log-Generation Rate 7

4 Network Distance and an Effective Spatial Candidate **7**
4.1 Decomposition of C_{ij} 7
4.2 Spatial Distance 7
4.3 Coarse-Graining and the Metric Tensor 8
4.4 Time-Like and Spatial Constructions 8

5 Flow and Vorticity **9**
5.1 Log-Flow 9
5.2 The Continuity Equation 9
5.3 Helmholtz Decomposition 10
5.4 Vorticity and Closure Density 10
5.5 Role-Separated Construction Summary 11

6 Closure and Persistent Structure **11**
6.1 What Is Closure 11
6.2 The Minimal Closure Is Triangular 11
6.3 Closure as a Dynamical Attractor 11
6.4 Closure Distance and Mass 12

7 Phase Structure **12**
7.1 The Control Parameter 13
7.2 Four Phases 13
7.3 Candidate Phase Interpretation of Effective Laws 13
7.4 Extreme Regimes as Phase Transitions 14

8 Discussion **14**

9 Conclusion **15**

A Notation and Symbol Reference **16**

1 Introduction

Most physical theories begin from assumed structure: a spacetime manifold, a Lagrangian density, or a symmetry group. In all such cases, the framework presupposes what it seeks to explain.

VED takes a different approach. Starting from the single statement *there is difference*, the framework proposes formal constructions and candidate interpretations for time, space, motion, and persistence. It does not establish that these physical targets follow necessarily from every instance of structural asymmetry.

The central object is the **causal log matrix** C_{ij}, defined on a discrete network of nodes. C_{ij} captures not the total history of causal influence between nodes, but the portion of that history that is selectively retained as persistent structure. From C_{ij}, a compact chain summarizes the proposed development:

$$\text{difference} \;\to\; C_{ij} \;\to\; \rho \;\to\; \tau \;\to\; d_{ij} \;\to\; J \;\to\; \omega \;\to\; L \tag{1}$$

See Fig. 1 for a visual summary.

The arrows in this compact display are navigation markers rather than one repeated relation. In the detailed construction, $C_{ij} \to \rho$ and the discrete $C_{ij} \to J_{ij}$ relation are definitions; $\rho \to \tau$ is an accumulation definition relative to the parameter t; $S_{ij} \to d_{ij}$ is a weighted-network construction; $J_{ij} \to \mathbf{J}(x)$ is an OPEN discrete-to-continuum bridge; $\mathbf{J} \to \omega$ is a continuum definition; and $\omega \to L$ is an effective proportional relation. Physical time, space, and persistence remain proposed interpretations rather than additional formal consequences of the arrow glyph.

This framework is not presented as a complete predictive theory, but as a minimal structural proposal for physical organization. It is a structural proposal: a demonstration that the minimum ingredients required for physical structure are considerably smaller than those assumed in current foundational approaches. The aim is coherence, not closure.

Organization. Section 2 states the minimal axiom and traces its immediate consequences. Section 3 introduces the causal log and a time-like accumulation quantity. Section 4 constructs network distance from the symmetric component of C_{ij}. Section 5 develops flow and vorticity from the antisymmetric component. Section 6 introduces a model-local minimal enclosure candidate. Section 7 establishes the phase structure of the network. Sections 8 and 9 discuss implications and open problems.

2 The Minimal Axiom

VED is founded on a single, irreducible statement:

There is difference.

This is not a metaphor or a philosophical preamble. It is a structural condition. A perfectly symmetric system produces no gradients, no flows, and no observable events. Without asymmetry, nothing happens. The overall structure is summarized in Fig. 1.

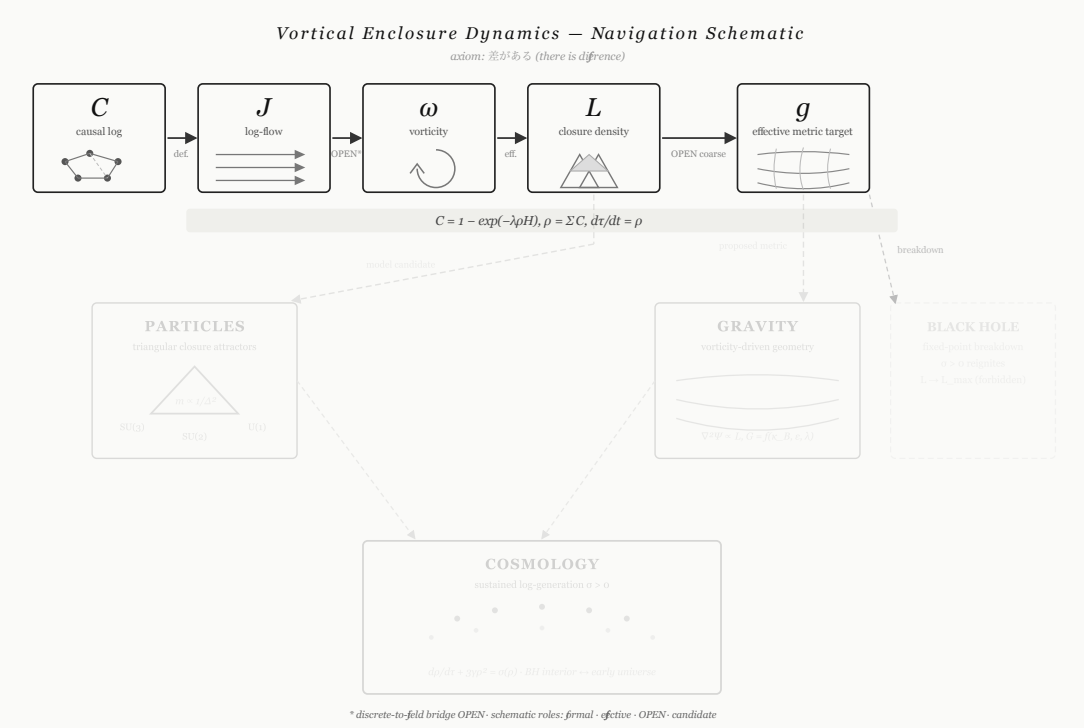


Figure 1: Historical compact chain of Vortical Enclosure Dynamics. Its stages include definitions, effective relations, open representation bridges, and candidate physical interpretations; its arrows therefore do not have one uniform derivational or generative meaning: C_{ij} and ρ support a time-like accumulation quantity τ, connectivity defines network distance d_{ij}, and the represented flow and vorticity motivate closure L.

Therefore:

Proposition 2.1. *An observable world requires the existence of difference.*

This is not an empirical claim; it is a structural necessity. If the world were completely symmetric, every state would be indistinguishable from every other, and the concept of observation would be undefined.

2.1 From Difference to Dynamics

In the local-field representation used in this section, the proposed chain is:

$$\text{difference} \;\Rightarrow\; \nabla\delta \neq 0 \;\Rightarrow\; J \sim -\nabla\delta \;\Rightarrow\; \text{flow} \;\Rightarrow\; \text{vortex} \;\Rightarrow\; \text{closure}$$

Here $\nabla\delta$ denotes the gradient of the represented difference field, so a local neighborhood / differential structure on which ∇ acts is already part of this representation. The relation $J \sim -\nabla\delta$ is an effective proportional flow relation; the symbol $\sim$ does not fix its normalization or supply a universal derivation. Accordingly, within this representation differences can support an effective directed-flow description. The stronger necessity and pre-geometric interpretations of the whole chain are separate claims, not consequences of these two expressions alone.

2.2 The Non-Closure Constraint

If a closure were complete — if all structural differences were eliminated — all gradients would vanish and dynamics would cease. This contradicts the minimal axiom. Therefore the complete-closure limit cannot be treated as an ordinary attainable state of the generated description:

$$L < L_{\max} \quad \text{at all times.} \tag{2}$$

In this volume we represent this non-closure constraint by the following effective closure barrier:

$$B(L) = -\kappa_B \log\left(1 - \frac{L}{L_{\max}}\right), \qquad L \to L_{\max} \;\Rightarrow\; B(L) \to \infty. \tag{3}$$

This term should be read as a phenomenological encoding of the fact that complete closure would erase the differences required for dynamics. It is not meant to be the final microscopic origin of non-closure. In Vol. 4, the same constraint is reinterpreted dynamically: the effective barrier may emerge from delayed rotational feedback, dissipation, and the instability of complete closure within a generated description.

Remark 2.2. The non-closure constraint has far-reaching consequences. In section 6, it provides an effective scaling for the mass spectrum of persistent structures. In Vol. 3, it is used to reinterpret the black hole center as an asymptotic closure limit rather than an ordinary singular object.

3 Causal Logs and Local Time-Like Accumulation

3.1 History and Structure

We distinguish two directed relations between nodes i and j:

- $H_{ij} \geq 0$: the **causal history** — total accumulated influence from i to j, irreversible and unfiltered.
- $C_{ij} \in [0, 1)$: the **causal log** — the selected portion of that history that becomes persistent structure. C_{ij} represents the strength of retained influence: not what happened, but what continues to matter.

Remark 3.1 (Source-local terminology). Throughout Vol. 1, C_{ij} denotes the causal-log matrix defined above. Some repository navigation records use the same glyph for a closure-degree or first-formal-carrier variable and describe the causal log separately. No identity map between those source-specific uses is asserted in this volume.

The fundamental distinction is:

$$H_{ij} \neq C_{ij}. \tag{4}$$

The world is not the sum of everything that occurred, but the structure of what was retained.

3.2 The Self-Referential Selection Equation

Selection is not externally imposed. It is generated by the structure itself. Define the **log density** at node i:

$$\rho_i = \sum_j C_{ij}. \tag{5}$$

The **causal log equation** is:

$$C_{ij} = 1 - \exp(-\lambda\, \rho_i\, H_{ij}), \tag{6}$$

where $\lambda > 0$ is the selection sensitivity, the sole external parameter.

Substituting equation (5) into equation (6) yields a self-referential fixed-point equation in ρ_i:

$$\rho_i = \sum_j \left[1 - \exp(-\lambda\, \rho_i\, H_{ij})\right].$$

Candidate stable fixed points ρ_i^*, if separately established, may be used as model candidates for persistent structure. A physical identification requires a separate stability and observation map.

For equation-role purposes, the displayed equations form a constitutive self-consistency relation and a proposed fixed-point target on node-indexed quantities. They are not, by themselves, a specified update map. Their display does not establish existence, uniqueness, convergence of an iteration, or stability of a solution; those properties require separate conditions.

Remark 3.2. The self-referential structure is not circular. It admits an iterative interpretation in which existing structure (ρ_i) governs how new history (H_{ij}) is selected into new structure (C_{ij}). No particular update schedule or convergence result is specified by that interpretation. Structure generates its own selectivity.

3.3 Local Time-Like Accumulation

The node-local quantity τ_i is not introduced as a background clock. It is defined relative to the derivative / ordering parameter t as follows:

Remark 3.3 (Derivative parameter). In this Vol. 1 formulation, t denotes the ordering / derivative parameter used to write the following relation and the dot derivatives below. This notation does not identify t with τ_i, with physical clock time, or with the effective external t used in IFGT.

$$\frac{d\tau_i}{dt} = \rho_i. \tag{7}$$

This equation is the source-defined node-local time-like accumulation relation relative to the derivative parameter t. Conditional on $\rho_i \geq 0$, it makes τ_i non-decreasing with respect to that

parameter. The equation alone does not identify τ_i with a physical clock reading or establish physical-time emergence.
The following mathematical consequences and interpretation limits are kept separate:

- Since $\rho_i \geq 0$ by construction, τ_i is monotonically non-decreasing with respect to t.
- Where $\rho_i = 0$, the defined quantity τ_i does not advance with respect to t.
- This monotonicity does not by itself establish state non-reconstructibility, thermodynamic irreversibility, or a physical arrow of time.

> *τ_i is a node-local accumulation / ordering quantity. Its interpretation as physical time remains proposed rather than established.*

3.4 The Log-Generation Rate

Taking the time derivative of ρ_i using equation (6) yields:

$$\dot{\rho}_i = \frac{\lambda\, \rho_i\, B_i}{1 - \lambda\, A_i}, \tag{8}$$

where $A_i = \sum_j e^{-\lambda \rho_i H_{ij}} H_{ij}$ and $B_i = \sum_j e^{-\lambda \rho_i H_{ij}} \dot{H}_{ij}$. The numerator $\lambda \rho_i B_i$ measures the rate of new history injection; the denominator $1 - \lambda A_i$ captures saturation by existing structure. This microscopic expression is proposed as an antecedent of the Vol. 1 continuum log-generation rate σ appearing in section 5. No microscopic-to-continuum projection or identity equation is fixed here.

4 Network Distance and an Effective Spatial Candidate

4.1 Decomposition of C_{ij}

The causal log matrix is generically asymmetric. We decompose it into symmetric and anti-symmetric parts:

$$S_{ij} = \tfrac{1}{2}(C_{ij} + C_{ji}), \qquad A_{ij} = \tfrac{1}{2}(C_{ij} - C_{ji}). \tag{9}$$

S_{ij} encodes mutual connectivity (the candidate spatial component); A_{ij} encodes directed flow (the temporal and causal component). These provide formal candidates for spatial and temporal / causal description from the same object C_{ij}; physical identification remains separate.

4.2 Spatial Distance

For a path $\gamma : i \to j$ through the network, define its **propagation strength**:

$$\tilde{S}_{ij} = \max_{\gamma : i \to j} \prod_{(ab) \in \gamma} S_{ab}.$$

The **distance** between nodes is:

$$d_{ij} = -\log \tilde{S}_{ij} = \min_{\gamma: i \to j} \sum_{(ab) \in \gamma} (-\log S_{ab}). \tag{10}$$

This is a shortest-path problem on a weighted network. The logarithm converts products to sums and ensures that distance accumulates additively along paths.

The construction is defined on a graph / network with an admissible path class and edge strengths for which the products and logarithms are defined. It is a weighted-path construction; physical-space interpretation is not part of the definition.

Proposition 4.1. *For reachable pairs with admissible strengths $0 < S_{ab} \leq 1$, d_{ij} is nonnegative and satisfies the triangle inequality inherited from path concatenation. Ordinary metric status additionally requires the applicable symmetry, finite-valuedness, and identity-of-indiscernibles conditions; a directed or disconnected network can instead yield an asymmetric or extended distance-like quantity, depending on the declared policy.*

The network distance measures resistance to influence propagation. It provides a formal candidate for effective spatial description, not a physical-space emergence theorem.

4.3 Coarse-Graining and the Metric Tensor

The source proposes a coarse-grained description in which nodes are grouped into spatial blocks and a smooth effective distance is represented by:

$$d(x, y)^2 \approx g_{\mu\nu}(x)\,(x^\mu - y^\mu)(x^\nu - y^\nu).$$

$$g_{\mu\nu} = \text{effective description of } d_{ij} \text{ under coarse-graining.} \tag{11}$$

The metric tensor $g_{\mu\nu}$ is not a fundamental field. It is proposed as an effective continuum description of the coarse-grained log-distance structure. The displayed endpoint relation is a partial bridge: an explicit reconstruction map, convergence argument, normalization, uniqueness result, and physical-distance identification are not supplied here. No coordinate system is assumed; coordinates label coarse-grained blocks. The displayed expression is an effective local continuum approximation in that target representation, not a uniqueness or reconstruction theorem and not a physical-space emergence result.

4.4 Time-Like and Spatial Constructions

The two formal constructions can be summarized as:

$$\frac{d\tau_i}{dt} = \rho_i \qquad \text{(time-like accumulation quantity)} \tag{12}$$

$$d_{ij} = -\log \tilde{S}_{ij} \qquad \text{(network spatial candidate)} \tag{13}$$

The two formal constructions use different aspects of the same underlying object C_{ij}. Their identification with physical time and physical space is not established by these equations alone.

5 Flow and Vorticity

5.1 Log-Flow

The antisymmetric component of C_{ij} defines the **log-flow**:

$$J_{ij} = C_{ij} - C_{ji} = 2A_{ij}. \tag{14}$$

J_{ij} represents the net directional transfer of causal influence. It satisfies $J_{ij} = -J_{ji}$ by construction. Motion arises from asymmetry: where $J_{ij} = 0$, no net transfer occurs; where $J_{ij} \neq 0$, directed propagation occurs.

5.2 The Continuity Equation

We use $\mathbf{J}(x)$ as a coarse-grained continuum representation of the discrete log-flow J_{ij}. This is an informal bridge: an explicit averaging, scaling, orientation, and discrete-to-vector-field map is not supplied here. The evolution of continuum log density ρ is written as:

$$\partial_t \rho + \nabla \cdot \mathbf{J} = \sigma, \tag{15}$$

Equation (15) is a balance / continuity relation within this continuum field representation. Because the source term may be nonzero, the display is not an unconditional conservation law and does not, by its balance form alone, establish a physical-time interpretation.

The continuum field ρ is associated with the node quantity ρ_i, but no interpolation, volume normalization, unit map, or complete function signature relating them is supplied here. Both continuum bridges remain open.

Here $\sigma \geq 0$ is the **log-generation rate**. σ is not an external source term. It is the rate at which the system fails to remain closed: wherever structure is incomplete, new logs are generated.

Remark 5.1 (Source-local symbol). Here σ denotes the Vol. 1 continuum log-generation rate. It is not identified by notation alone with the microscopic $\dot{\rho}_i$ expression or with IFGT's separately constructed generation term σ; those translations remain unspecified.

This is a balance relation rather than an unconditional conservation law. Within the stated continuum model, $\sigma = 0$ marks the source-free regime associated with the classical fixed-point

phase (section 7). This formal regime distinction does not by itself establish whether physical conservation laws are fundamental or emergent.

$$\sigma > 0 \qquad \text{(positive source regime)}. \tag{16}$$

Within this source-local continuum model, $\sigma > 0$ denotes positive log generation and is proposed as an indicator of accumulation / temporal ordering. Neither direction of an equivalence with physical-time generation is established by the balance equation.

5.3 Helmholtz Decomposition

The flow field admits a Helmholtz decomposition:

$$\mathbf{J} = \mathbf{J}_{\text{rot}} + \mathbf{J}_{\text{pot}},$$

where $\nabla \cdot \mathbf{J}_{\text{rot}} = 0$ and $\nabla \times \mathbf{J}_{\text{pot}} = 0$. The solenoidal component $\mathbf{J}_{\text{rot}}$ carries vorticity and is used in the proposed enclosure mechanism. Persistence and stability require additional dynamical analysis. The irrotational component $\mathbf{J}_{\text{pot}}$ carries the divergence associated with the log-generation rate σ.

5.4 Vorticity and Closure Density

Define the **vorticity** of the log-flow:

$$\omega = \nabla \times \mathbf{J}. \tag{17}$$

This is a definition, and hence a derived construction, in a continuum vector-field setting in which the curl operator is meaningful. It defines the vorticity of the represented log-flow; nonzero ω alone is not a stability or persistence theorem.

Vorticity measures local circulation. In the subsequent enclosure interpretation of the model, $\omega = 0$ denotes an irrotational represented flow. When $\omega \neq 0$, circulation is present and flow lines may close; this is a candidate structural route to enclosure, not a theorem of persistence.

The local intensity of vortical structure defines the **closure density**:

$$L \sim |\omega|. \tag{18}$$

This is an effective proportional identification of a scalar closure-intensity measure in the present representation. The magnitude convention and proportionality normalization are not fixed here, and the relation is not a derived universal identity.

Remark 5.2 (Carrier). The L in this section is a scalar local closure-density / vortical-intensity quantity. The edge quantity L_{ab} and coarse tensor $L_{\mu\nu}$ introduced in Vol. 3 are distinct carrier-specific objects; shared lettering does not define a map among them.

L is used in Vols. 2–3 as part of candidate structural interpretations of particle-like and gravitational behavior; those physical identifications are separate from the definition above.

5.5 Role-Separated Construction Summary

The following historical compact sequence summarizes the route from asymmetry to the scalar closure measure:

$$C_{ij} \xrightarrow{J=C-C^T} J \xrightarrow{\omega=\nabla\times J} \omega \xrightarrow{L\sim|\omega|} L.$$

Its first edge is a discrete definition, followed by an OPEN discrete-to-continuum representation bridge before the curl definition; the last edge is an effective proportional relation. The arrows therefore do not share one logical status, and the sequence alone does not establish persistent physical enclosure.

6 Closure and Persistent Structure

6.1 What Is Closure

A **closure** is a configuration in which causal flow circulates along a loop, the loop reinforces itself through the self-referential selection equation equation (6), and the resulting pattern is dynamically sustained. A closure is not a boundary, but a circulation that sustains itself.

Closure is proposed as a mechanism that can support flow trapping and maintained gradients rather than dissipation. The sustained behavior in the model definition above is a target, not a persistence guarantee from the self-consistency equation; its realization requires dynamical stability analysis.

6.2 The Minimal Closure Is Triangular

Within the finite simple-loop closure class considered here, the first candidate nontrivial enclosure involves three nodes:

- **Two nodes**: the only cycle $i \to j \to i$ is purely oscillatory, with no internal degrees of freedom and no stable phase structure.
- **Three nodes**: the cycle $i \to j \to k \to i$ admits an oriented loop with internal area. The three edges span a two-dimensional phase space, allowing stable phase relationships.
- **Four or more**: in the stated finite decomposition picture, these cycles are treated as compositions of triangular sub-loops.

> *Within the stated finite simple-loop closure class, the triangle is the minimal candidate enclosure. Unrestricted uniqueness is not claimed.*

6.3 Closure as a Dynamical Attractor

Define the **closure energy** of a configuration A:

$$E(A) = \alpha \left| \sum_{i \in A} \vec{e}_i \right|^2 - \beta\, \Phi(A) + \gamma |A| + B(L(A)), \tag{19}$$

where $\vec{e}_i$ are internal flux vectors derived from C_{ij}, $\Phi(A) = \sum_{i<j<k\in A} \phi_{ijk}$ is the non-degeneracy measure, and $B(L)$ is the effective non-closure barrier of equation (3). Thus $E(A)$ is the model energy / scoring functional used to compare configurations in this closure model; the definition of the functional does not by itself identify one of its minima with an observed particle.

Definition 6.1. *A **particle** is a local minimum of the closure energy $E(A)$.*

Here "particle" is model terminology for a closure-energy minimum. Such a minimum is a particle-like structural candidate; identification with an observed particle requires a separate stability and observational map.

6.4 Closure Distance and Mass

The **closure distance** of a stable configuration:

$$\Delta_k = L_{\max} - L_k \tag{20}$$

measures how close the configuration lies to the complete-closure limit. The curvature of E at the minimum gives the effective mass. Within the phenomenological barrier approximation this yields:

$$m_k \propto \left.\frac{d^2E}{dL^2}\right|_{L_k} \sim \frac{\kappa_B}{\Delta_k^2}. \tag{21}$$

The displayed scaling is retained only within that phenomenological barrier-based scope. It is not a microscopic derivation or an identification of m_k with an observed mass without additional interpretation.

Here κ_B is an effective parameter of the projected closure-energy description. Vol. 4 gives a more dynamical interpretation in which the apparent barrier can arise from rotational feedback and dissipation rather than from a fundamental external potential.

Within the phenomenological scaling, larger mass-like values occur close to the complete-closure limit. The world is made of incomplete closures.

This minimal structure is illustrated in Fig. 3.

7 Phase Structure

The dynamics of the causal log network do not produce uniform behavior. Depending on the magnitude of the effective coupling, the network operates in qualitatively different regimes. Physical laws are properties of a specific phase, not universal principles valid across all parameter regimes.

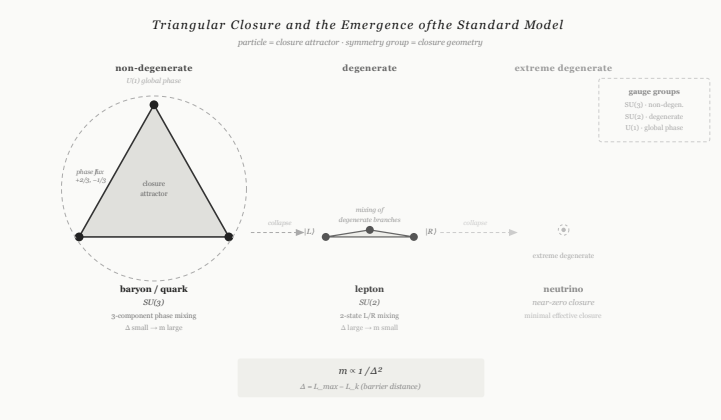


Figure 3: Triangular closure types and their proposed correspondence to particle-like families. The three configurations illustrate non-degenerate, degenerate, and extreme degenerate closures as candidate enclosures in the stated model, forming the basis for higher-order symmetries explored in subsequent work.

7.1 The Control Parameter

The key parameter is:

$$\alpha_i = \lambda\, h_i, \qquad h_i = \sum_j H_{ij}, \tag{22}$$

where λ is the selection sensitivity and h_i is the total incoming causal history at node i. The value $\alpha_i = 1$ marks the proposed threshold in this model's phase picture. Self-sustained log density and structure formation above it are candidate behaviors, not consequences established here: fixed-point existence, update semantics, and dynamical stability remain open. The proposed phase structure is shown in Fig. 2.

7.2 Four Phases

Phase	Condition	Character
Subcritical	$\alpha < 1$	$\rho \approx 0$; no time, no structure
Critical	$\alpha = 1$	Maximum susceptibility; bifurcation point
Classical-active	$\alpha \sim 1$	Candidate stable closures and effective laws
Saturated	$\alpha \gg 1$	$C_{ij} \to 1$; differences vanish

Table 1: Phase structure of the causal log network.

7.3 Candidate Phase Interpretation of Effective Laws

Within the proposed classical-active interpretation, the following structures are candidate stable features:

- **Conservation laws**: $\sigma \approx 0$, so $\partial_t \rho + \nabla \cdot \mathbf{J} \approx 0$.

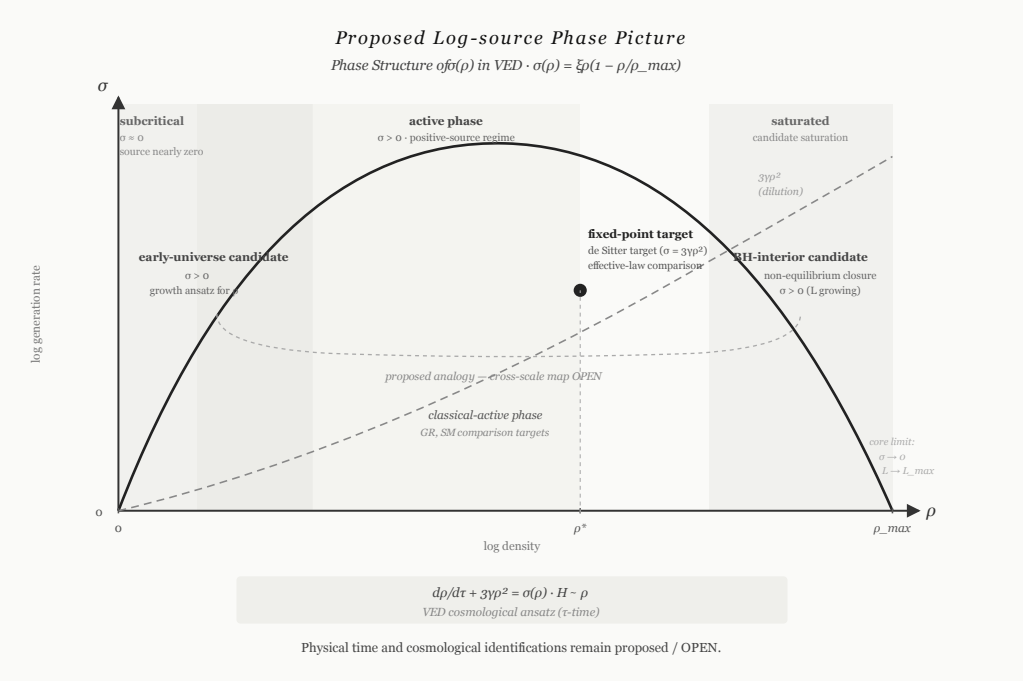


Figure 2: Phase structure as a function of density ρ. The system transitions across subcritical, critical, classical-active, and saturated regimes. Effective-law behavior is a candidate phase interpretation.

- **Gauge symmetries**: closure configurations stabilize with well-defined internal degrees of freedom (Vol. 2).
- **Effective geometry**: under the proposed partial coarse correspondence, d_{ij} is represented approximately by a smooth metric tensor $g_{\mu\nu}$ (Vol. 3); reconstruction, convergence, and uniqueness of this bridge remain open.

Conservation, symmetry, and geometry are modeled as properties of the classical-active phase rather than as axioms. Their physical phase identification remains open.

7.4 Extreme Regimes as Phase Transitions

The framework proposes model interpretations of extreme regimes:

- **Big Bang candidate**: a proposed transition from subcritical to active phase as α crosses the critical threshold.
- **Black-hole-interior candidate**: a proposed transition from classical-active to non-equilibrium behavior where fixed-point conditions break down and $\sigma > 0$ is used (Vol. 3).

8 Discussion

VED investigates whether structures used as foundational primitives in physics — spacetime, fields, and symmetry groups — can receive constructions or candidate interpretations from a

self-referential process on a discrete causal network. The present work should be understood as a structural proposal, not a finalized physical theory. Several aspects of this proposal merit explicit discussion.

Relation to existing frameworks. VED does not quantize general relativity, nor does it discretize quantum field theory. Its closest structural relatives are causal set theory [1, 2], in which spacetime is replaced by a partial order on discrete events, and relational quantum mechanics [3], which locates physical quantities in relations between systems rather than in absolute values. VED differs from both in proposing spatial and temporal constructions from a single self-referential equation without postulating an underlying causal order. Their physical identification remains open.

Status of conservation laws. The continuity equation equation (15) does not impose conservation. Conservation ($\sigma = 0$) is a special case that holds in the classical-active phase. This is a departure from standard formulations, where conservation is an axiom or a consequence of symmetry (Noether's theorem). In VED, Noether's theorem applies within the classical phase as an effective statement; it does not hold in general.

The Lagrangian as an effective description. An early version of this framework attempted a Lagrangian formulation. This was abandoned for two reasons: (i) the Lagrangian principle $\delta S = 0$ uses a time parameter t, while VED defines an internal time-like accumulation quantity τ whose physical clock interpretation remains proposed; (ii) the non-zero log-generation rate σ is structurally incompatible with a variational principle that requires conservation. The Lagrangian formalism is proposed as an effective comparison within the classical-active phase, where $\sigma \approx 0$; a recovery theorem is not given here.

Open problems. The framework presented here is structurally complete but quantitatively open. Key open problems include:

1. Derivation of the specific value of λ from first principles.
2. Proof of existence and uniqueness of fixed points for generic H_{ij}.
3. Quantitative derivation of mass ratios from Δ_k (Vol. 2).
4. Exact (non-linear) Schwarzschild solution from the fixed-point equation (Vol. 3).
5. Microscopic computation of Newton's constant G from λ, κ_B, and network density (Vol. 3).

9 Conclusion

We have presented a structural framework proposing time-like, spatial, and persistence constructions from a single axiom: *there is difference*.

The central organizing summary is the compact navigation mnemonic equation (1), in which the causal log matrix C_{ij}, defined by the self-referential equation equation (6), is used to construct or motivate: a node-local accumulation quantity τ (equation (7)), network distance d_{ij}

(equation (10)), log-flow J (equation (14)), vorticity ω (equation (17)), closure density L (equation (18)), and a phenomenological mass-like scaling for candidate persistent structures (equation (21)). Physical identifications remain separate from these formal constructions. The edges in the mnemonic include definitions, a weighted-network construction, an effective relation, and an OPEN discrete-to-continuum bridge; they do not form one uniform physical-generation rule.

The system exhibits four phases (table 1). The observable universe is compared with the classical-active phase, in which conservation laws and effective geometry are proposed as stable features. That physical identification and its universality remain open.

> *The world is not given. It is constructed. It persists because it never fully closes.*

Volumes 2 and 3 develop the consequences of this framework for the Standard Model and for gravity and cosmology, respectively.

A Notation and Symbol Reference

The following table summarizes the principal symbols used in this volume.

Symbol	Definition	Introduced
H_{ij}	Causal history ($i \to j$ influence strength)	section 3
C_{ij}	Causal log (selected structure)	equation (
ρ_i	Log density $= \sum_j C_{ij}$	equation (
λ	Selection sensitivity (sole external parameter)	equation (
τ_i	Node-local time-like accumulation quantity ($d\tau_i/dt = \rho_i$)	equation (
t	Vol. 1 ordering / derivative parameter (not an asserted clock or cross-source identity)	section 3
S_{ij}	Symmetric component of C_{ij} (spatial)	equation (
A_{ij}	Antisymmetric component of C_{ij} (causal)	equation (
$\tilde{S}_{ij}$	Maximum-path propagation strength	section 4
d_{ij}	Network distance $= -\log \tilde{S}_{ij}$	equation (
$g_{\mu\nu}$	Effective metric candidate (physical interpretation open)	equation (
J_{ij}	Log-flow $= C_{ij} - C_{ji}$	equation (
σ	Vol. 1 continuum log-generation rate	equation (
ω	Vorticity $= \nabla \times \mathbf{J}$	equation (
L	Scalar closure-density / intensity quantity $\sim \lvert\omega\rvert$	equation (
$L_{\max}$	Complete-closure limit	equation (
$B(L)$	Effective non-closure barrier	equation (
κ_B	Effective barrier strength	equation (
$E(A)$	Closure energy of configuration A	equation (
Δ_k	Closure distance $= L_{\max} - L_k$	equation (
m_k	Effective mass scaling $\propto \kappa_B/\Delta_k^2$	equation (
α_i	Control parameter $= \lambda h_i$	equation (

Table 2: Symbol reference for Vol. 1.

References

[1] Luca Bombelli, Joohan Lee, David Meyer, and Rafael D. Sorkin. Space-time as a causal set. *Physical Review Letters*, 59(5):521–524, 1987.

[2] Rafael D. Sorkin. Causal sets: Discrete gravity. *Lectures on Quantum Gravity*, pages 305–327, 2005.

[3] Carlo Rovelli. Relational quantum mechanics. *International Journal of Theoretical Physics*, 35(8):1637–1678, 1996.